



$$\mathcal{L}(\varphi, \partial_m \varphi, x)$$

$$= \left| \frac{\partial x'}{\partial x} \right| \mathcal{L}(\varphi', \partial'_m \varphi', x')$$

absolute value
of the determinant
of the Jacobian
matrix

INGREDIENTS

① Jacobian term:

transformation of coordinates:

$$x'^m = x^m + \delta x^m$$

$$\text{Hence, } \frac{\partial x'^m}{\partial x^p} = \underbrace{\frac{\partial x^m}{\partial x^p}}_{\delta_p^m} + \underbrace{\frac{\partial \delta x^m}{\partial x^p}}_{\partial_p \delta x^m} \quad \left(\partial_p = \frac{\partial}{\partial x^p} \right)$$

$$\text{Then } \frac{\partial x'^m}{\partial x^p} = \underbrace{\delta_p^m}_{\text{zeroth order}} + \underbrace{\partial_p \delta x^m}_{\text{1st order in } \delta}$$

$\Rightarrow$ Jacobian matrix has
the structure: $1 + \epsilon$
 $\uparrow$ correction $\ll 1$

Hence:

$$\det(1 + \epsilon) = \exp[\text{Tr}[\ln(1 + \epsilon)]]$$

$$\text{1st order } \approx \exp[\text{Tr}[\epsilon]]$$

$$\text{1st order} = 1 + \text{Tr}[\epsilon]$$

$$\text{So, } \det\left(\frac{\partial x'}{\partial x}\right) = 1 + \partial_m \delta x^m$$

Einstein summation convention (over repeated indices)

But since $\partial_m \delta x^m$ is a first order correction,
then its absolute value is smaller than one

$$\Rightarrow \left| \det\left(\frac{\partial x'}{\partial x}\right) \right| \approx \det\left(\frac{\partial x'}{\partial x}\right) = 1 + \partial_m \delta x^m$$

② the $\phi'(x')$ term:

we follow that:

$$\phi'(x') = \phi'(x + \delta x)$$

Expansion
at first
order

$$\Rightarrow \underbrace{\phi'(x)}_{\text{zeroth order in } \delta} +$$

coordinate
transfor.

$$\sum_m \left(\frac{\partial \phi'}{\partial x^{m'}} \right)_{x'=x} \delta x^m$$

1st order term in " δ "

1st order in " δ "

can be approximated by a zeroth order term
coming from another expansion

But : $\left(\frac{\partial \phi'}{\partial x'^M} \right) = \underbrace{\left(\frac{\partial \phi}{\partial x^M} \right)}_{\text{zeroth order terms}} + \text{1st order correction terms in "}\delta\text{"}$

Hence : $\phi'(x) = \phi(x) + \sum_M (\partial_M \phi) \delta x^M$

Moreover :

★ Definition : we introduce the variation of the fields $\delta \phi(x)$ AT THE SAME POINT x by : $\delta \phi(x) = \phi'(x) - \phi(x)$

finally :

$$\phi'(x') = \phi(x) + \delta \phi(x) + (\partial_M \phi) \delta x^M$$

③ the $(\partial_n \phi')(x')$ term :

Starting point :

$$\frac{\partial \phi'}{\partial x'^M} = \frac{\partial}{\partial x'^M} \left\{ \phi(x) + \delta \phi(x) + (\partial_P \phi) \delta x^P \right\}$$

$$= \underbrace{\frac{\partial x^N}{\partial x'^M}}_{\text{inverse of the jacobian matrix}} \frac{\partial}{\partial x^N} \left\{ \phi(x) + \delta \phi(x) + (\partial_P \phi) \delta x^P \right\}$$

$\frac{\partial x^N}{\partial x'^M} = \delta_N^M + \partial_N \delta x^M$
 $(1 + \epsilon)^{-1} \simeq 1 - \epsilon$
 $\text{"}1 + \epsilon\text{"}$

$$\Rightarrow \delta_N^M - \partial_N \delta x^M$$

$$= (\underbrace{\delta_N^M}_{\text{0th order}} - \underbrace{\partial_N \delta x^M}_{\text{1st order}}) \left\{ \underbrace{\partial_n \phi}_{\text{zeroth order}} + \underbrace{\partial_n \delta \phi}_{\text{first order}} + \underbrace{\partial_n [(\partial_P \phi) \delta x^P]}_{\text{first order}} \right\}$$

$(\partial_n \partial_P \phi) \delta x^P + (\partial_P \phi) (\partial_n \delta x^P)$

$$\begin{aligned} & \left[\begin{aligned} & \xrightarrow{\text{first order in "}\delta\text{"}} \partial_m \phi + \partial_m \delta\phi + (\partial_m \partial_p \phi) \delta x^p \\ & + \cancel{(\partial_p \phi)(\partial_m \delta x^p)} \end{aligned} \right] \quad \delta x^\mu \text{ contribution} \\ & \star \Rightarrow \left[- \cancel{(\partial_m \delta x^\mu)(\partial_n \phi)} \right] - \partial_m \delta x^\mu \text{ contribution} \end{aligned}$$

finally :
$$\partial'_m \phi' = \partial_m \phi + \partial_m \delta\phi + (\partial_m \partial_p \phi) \delta x^p$$

④ The $\mathcal{L}(\phi', \partial'\phi'; x')$ term :

the starting point is :

$$\mathcal{L}(\underbrace{\phi + \delta\phi + (\partial_m \phi) \delta x^m}_{\in}, \underbrace{\partial_m \phi + \partial_m \delta\phi + (\partial_m \partial_p \phi) \delta x^p}_{1^{\text{st}} \text{ order "}\epsilon\text{"}}, \underbrace{x^m + \delta x^m}_{\in \text{ term}})$$

$$\begin{aligned} & \xrightarrow{1^{\text{st}} \text{ order } \mathcal{L}(\phi, \partial\phi, x)} \mathcal{L}(\phi, \partial_m \phi, x) + (\delta\phi + (\partial_m \phi) \delta x^m) \frac{\partial \mathcal{L}}{\partial \phi} \\ & + (\partial_m \delta\phi + (\partial_m \partial_p \phi) \delta x^p) \frac{\partial \mathcal{L}}{\partial (\partial_m \phi)} + \delta x^m \frac{\partial \mathcal{L}}{\partial x^m} \\ & + \text{Second order terms} \end{aligned}$$

However, the total derivative of the Lagrangian density reads :

$$\partial_p \mathcal{L} = (\partial_p \phi) \frac{\partial \mathcal{L}}{\partial \phi} + (\partial_p \partial_m \phi) \frac{\partial \mathcal{L}}{\partial (\partial_m \phi)} + \frac{\partial \mathcal{L}}{\partial x^p}$$

$p \rightarrow m$
 $m \rightarrow p$

Then :

$$\begin{aligned} \mathcal{L}(\phi', \partial'\phi', x') &= \mathcal{L}(\phi, \partial\phi, x) + \delta x^m \underbrace{x \left[(\partial_m \phi) \frac{\partial \mathcal{L}}{\partial \phi} + (\partial_p \partial_m \phi) \frac{\partial \mathcal{L}}{\partial (\partial_p \phi)} + \frac{\partial \mathcal{L}}{\partial x^m} \right]}_{\partial_m \mathcal{L}} \\ &+ \delta\phi \frac{\partial \mathcal{L}}{\partial \phi} + (\partial_m \delta\phi) \frac{\partial \mathcal{L}}{\partial (\partial_m \phi)} \end{aligned}$$

We have found:

$$\mathcal{L}(\phi', \partial\phi', x') = \mathcal{L}(\phi, \partial\phi, x) + (\partial_m \mathcal{L}) \delta x^m + \delta\phi \frac{\partial \mathcal{L}}{\partial \phi} + (\partial_m \delta\phi) \frac{\partial \mathcal{L}}{\partial(\partial_m \phi)}$$

Symmetry Condition

$$\begin{aligned} \mathcal{L}(\phi, \partial\phi, x) &= \left| \frac{\partial x'}{\partial x} \right| \mathcal{L}(\phi', \partial'\phi', x') \\ &= (1 + \partial_m \delta x^m) \left[\underbrace{\mathcal{L}(\phi, \partial\phi, x)}_{\mathcal{L}} + (\partial_p \mathcal{L}) \delta x^p + \delta\phi \frac{\partial \mathcal{L}}{\partial \phi} + (\partial_p \delta\phi) \frac{\partial \mathcal{L}}{\partial(\partial_p \phi)} \right] \\ &\stackrel{\uparrow}{=} \underset{\text{1st order}}{\mathcal{L}} + \delta\phi \frac{\partial \mathcal{L}}{\partial \phi} + (\partial_p \delta\phi) \frac{\partial \mathcal{L}}{\partial(\partial_p \phi)} + \underbrace{(\partial_p \mathcal{L}) \delta x^p}_{\text{comes from "1"}} \\ &\quad + \underbrace{(\partial_m \delta x^m) \mathcal{L}}_{\partial_p (\mathcal{L} \delta x^p)} \quad \left] \text{ comes from " } \partial_m \delta x^m \text{ " } \right. \end{aligned}$$

So, at first order in " δ "

$$\mathcal{L} = \mathcal{L} + \delta\phi \frac{\partial \mathcal{L}}{\partial \phi} + (\partial_p \delta\phi) \frac{\partial \mathcal{L}}{\partial(\partial_p \phi)} + \partial_p (\mathcal{L} \delta x^p)$$

↪ $A = 0$

But:

$$\underbrace{(\partial_p \delta\phi)}_{U'} \underbrace{\frac{\partial \mathcal{L}}{\partial(\partial_p \phi)}}_V = \underbrace{\partial_p}_{(U V)'} \left[\underbrace{\delta\phi}_{(U V)'} \underbrace{\frac{\partial \mathcal{L}}{\partial(\partial_p \phi)}}_{(U V)'} \right] - \delta\phi \times \partial_p \frac{\partial \mathcal{L}}{\partial(\partial_p \phi)}$$

Then:

$$A = \delta\phi \frac{\partial \mathcal{L}}{\partial \phi} - \delta\phi \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} + \partial_\mu \left(\delta\phi \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) + \partial_\mu (\mathcal{L} \delta x^\mu) = 0$$

$$\Leftrightarrow \delta\phi \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right] = -\partial_\mu \left(\mathcal{L} \delta x^\mu + \delta\phi \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right)$$

involves the
coordinate transformations

involves the field
modification at the same
point

• c) the Noether Theorem:

Using the same assumptions as before. Moreover, $\phi(x)$ is a solution of the equation of motion, so it satisfies

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = 0$$

$$\Rightarrow \partial_\mu \left(\mathcal{L} \delta x^\mu + \delta\phi \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) = 0$$

Noether Theorem: For any trajectory of a system having a symmetry,

there is a current J^μ which is conserved: $\partial_\mu J^\mu = 0$

$$\text{with } J^\mu = \mathcal{L} \delta x^\mu + \delta\phi \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)}$$

J^μ : Noether current

Comments:

① Properties of the J^M components.

With $\left\{ \begin{array}{l} J^0: \text{temporal component} \\ J^i: \text{spatial components} \end{array} \right. \quad i \in [1, D-1]$

$$\text{then } \boxed{\partial_0 J^0 + \partial_i J^i} = 0$$

equation of conservation
" $\partial_t \rho + \text{div } \vec{J} = 0$ "

Def: the charge Q associated to the Noether current is:

$$Q = \int_{\tilde{\Sigma}_{0,1}} d^{D-1}x J^0(x)$$

integration over all of the space.
NOT the time

Q should in principal depends on time, however:

$$\frac{dQ}{dt} = \int_{\tilde{\Sigma}_{0,1}} d^{D-1}x \partial_0 J^0$$

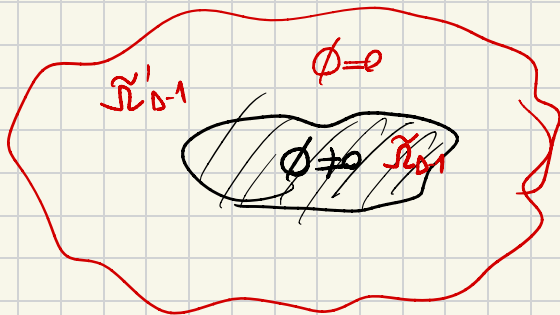
$$= - \int_{\tilde{\Sigma}_{0,1}} d^{D-1}x \partial_i J^i$$

Ostrogradski-Green theorem
 $\Rightarrow$ flux integration over the boundary of $\tilde{\Sigma}_{0,1}$

In physics, there are 2 possibilities:

1) $\phi \neq 0$ on a closed domain of space

$\Rightarrow$ we can choose the boundary to make the integral vanish



$$\int_{\tilde{\Sigma}_{0,1}} d^{D-1}x \partial_i J^i = \int_{\tilde{\Sigma}'_{0,1}} d^{D-1}x \partial_i J^i$$

2° ϕ decreases "fast enough" at infinity so that

$$\frac{\partial Q}{\partial t} = 0 \quad \Rightarrow \quad Q \text{ is a constant}$$

② Unicity of the Noether current :

The initial relation defines a structure for J^M

But J^M is defined from its divergence. So, show that :

$$J'^M = J^M + \partial_M V^{MM} \quad \text{where } V^{MM} = -V^{MM}$$

is also a Noether current (it satisfies the conservation law)

③ how to determine the Noether current : 2 ways

1st : use the expansion of J^M

2nd : compute $\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)}$ and see if it can be put into the form of a divergence term